Angles in Polygons

Name _____ **Date** _____ No calculator Version 1

KEY WORDS polygon · regular polygon · interior angle · exterior angle · angle sum · pentagon · hexagon · octagon

A The sum of the interior angles ●○○ — ANSWER ON THE SHEET

Remember: A polygon with n sides can be cut into $n - 2$ triangles, so its interior angles add to $180 \times (n - 2)$ degrees.

1. Complete the table.

Polygon	Number of sides	Number of triangles	Sum of the interior angles
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5		
Hexagon	6		
Octagon	8		

2. Find the sum of the interior angles of a polygon with:

a) 7 sides _____ **b)** 9 sides _____ **c)** 12 sides _____

d) 20 sides _____ **e)** 15 sides _____ **f)** 102 sides _____

B Finding a missing angle ●●○ — WORK IN YOUR BOOK

3. Work out the sum of the angles first.

a) The angles of a quadrilateral are 90°, 85°, 110° and x. Find x.

b) The angles of a pentagon are 100°, 120°, 95°, 110° and x. Find x.

c) The angles of a hexagon are 130°, 110°, 125°, 140°, 95° and x. Find x.

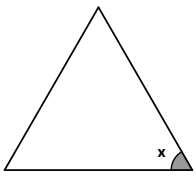
d) The angles of a pentagon are 108°, 108°, 108°, 108° and x. Find x.

C Regular polygons ●●○ — WORK IN YOUR BOOK

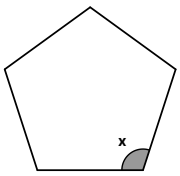
Remember: In a regular polygon all the sides are equal and all the angles are equal. So each interior angle is the angle sum divided by the number of sides.

4. Each shape is a regular polygon. Find the size of the angle marked x.

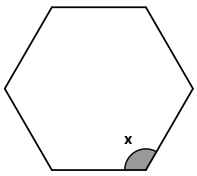
a)



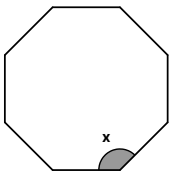
b)



c)



d)



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5. Find the size of each interior angle of:

a) A regular polygon with 9 sides

b) A regular polygon with 10 sides

c) A regular polygon with 12 sides

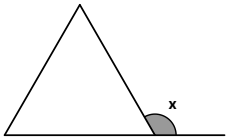
d) A regular polygon with 20 sides

D Exterior angles ●●○ WORK IN YOUR BOOK

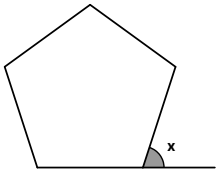
> **Remember:** The exterior angles of any polygon add to 360°. At each corner, the interior angle and the exterior angle add to 180°.

6. Each shape is a regular polygon with one side made longer. Find the size of the exterior angle marked x.

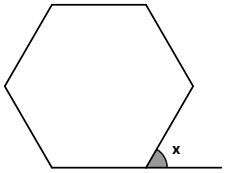
a)



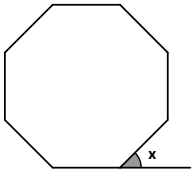
b)



c)



d)



7. Find the size of each exterior angle of:

a) A regular polygon with 9 sides

b) A regular polygon with 12 sides

c) A regular polygon with 20 sides

d) A regular polygon with 36 sides

8. Answer each question.

a) An interior angle of a polygon is 110°. Find the exterior angle at the same corner.

b) An exterior angle of a polygon is 45°. Find the interior angle at the same corner.

c) An interior angle of a polygon is 162°. Find the exterior angle at the same corner.

E How many sides? ●●● WORK IN YOUR BOOK

> **Remember:** The exterior angles add to 360°, so the number of sides of a regular polygon is 360 divided by one exterior angle.

9. How many sides does each regular polygon have?

a) Each exterior angle is 60°.

b) Each exterior angle is 40°.

c) Each exterior angle is 30°.

d) Each exterior angle is 72°.

e) Each interior angle is 120°.

f) Each interior angle is 135°.

g) Each interior angle is 150°.

h) Each interior angle is 108°.

★ Extension FOR EARLY FINISHERS

10. Can a regular polygon have interior angles of 100°? Show your working.

11. The interior angles of a polygon add to 1,260°. How many sides does it have?

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Angles in Polygons

Name _____ **Date** _____ No calculator Version 2

KEY WORDS polygon · regular polygon · interior angle · exterior angle · angle sum · pentagon · hexagon · octagon

A The sum of the interior angles ●○○ — ANSWER ON THE SHEET

Remember: A polygon with n sides can be cut into $n - 2$ triangles, so its interior angles add to $180 \times (n - 2)$ degrees.

1. Complete the table.

Polygon	Number of sides	Number of triangles	Sum of the interior angles
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5		
Hexagon	6		
Heptagon	7		

2. Find the sum of the interior angles of a polygon with:

a) 8 sides _____ **b)** 10 sides _____ **c)** 12 sides _____

d) 20 sides _____ **e)** 18 sides _____ **f)** 52 sides _____

B Finding a missing angle ●●○ — WORK IN YOUR BOOK

3. Work out the sum of the angles first.

a) The angles of a quadrilateral are 90°, 95°, 105° and x. Find x.

b) The angles of a pentagon are 105°, 115°, 100°, 120° and x. Find x.

c) The angles of a hexagon are 120°, 115°, 130°, 135°, 100° and x. Find x.

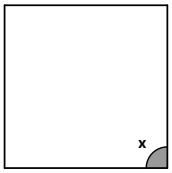
d) The angles of a hexagon are 120°, 120°, 120°, 120°, 120° and x. Find x.

C Regular polygons ●●○ — WORK IN YOUR BOOK

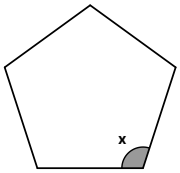
Remember: In a regular polygon all the sides are equal and all the angles are equal. So each interior angle is the angle sum divided by the number of sides.

4. Each shape is a regular polygon. Find the size of the angle marked x.

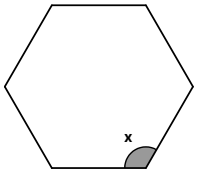
a)



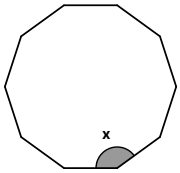
b)



c)



d)



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5. Find the size of each interior angle of:

a) A regular polygon with 8 sides

b) A regular polygon with 9 sides

c) A regular polygon with 12 sides

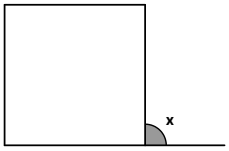
d) A regular polygon with 18 sides

D Exterior angles ●●○ WORK IN YOUR BOOK

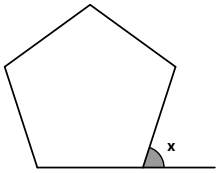
Remember: The exterior angles of any polygon add to 360°. At each corner, the interior angle and the exterior angle add to 180°.

6. Each shape is a regular polygon with one side made longer. Find the size of the exterior angle marked x.

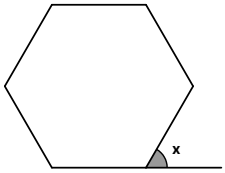
a)



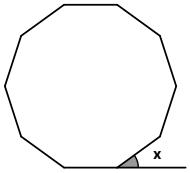
b)



c)



d)



7. Find the size of each exterior angle of:

a) A regular polygon with 10 sides

b) A regular polygon with 12 sides

c) A regular polygon with 18 sides

d) A regular polygon with 30 sides

8. Answer each question.

a) An interior angle of a polygon is 125°. Find the exterior angle at the same corner.

b) An exterior angle of a polygon is 40°. Find the interior angle at the same corner.

c) An interior angle of a polygon is 156°. Find the exterior angle at the same corner.

E How many sides? ●●● WORK IN YOUR BOOK

Remember: The exterior angles add to 360°, so the number of sides of a regular polygon is 360 divided by one exterior angle.

9. How many sides does each regular polygon have?

a) Each exterior angle is 90°.

b) Each exterior angle is 45°.

c) Each exterior angle is 36°.

d) Each exterior angle is 24°.

e) Each interior angle is 120°.

f) Each interior angle is 140°.

g) Each interior angle is 144°.

h) Each interior angle is 160°.

★ Extension FOR EARLY FINISHERS

10. Can a regular polygon have interior angles of 130°? Show your working.

11. The interior angles of a polygon add to 1,620°. How many sides does it have?

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